Python syntax cheat sheet

Python Lab cheat sheet. Snapshot 17 October 2026. This is a study aid, not a specification. Confirm details in the Sources section of Python Lab and on the official pages linked there.

Program shape

- Indentation groups a block. Use four spaces. Do not mix tabs and spaces.
- Comments start with #. A string literal is not a comment.
- print(value) writes text. The call happens because of the parentheses.
- A module runs top to bottom. if __name__ == '__main__' guards script-only code.

Names and built-in types

- Assignment binds a name to an object. It does not copy the object.
- int has arbitrary precision. float is a binary floating-point number.
- str is Unicode text. bytes is raw octets. They are not interchangeable.
- bool is a subclass of int. None is the only value of NoneType.
- list and dict are mutable. tuple, str, and frozenset are immutable.
- A trailing comma makes a one-item tuple: (1,).

Control flow and functions

- if / elif / else, while, for, break, continue, and pass.
- for walks any iterable. range(n) yields 0 through n-1.
- match / case is structural pattern matching, added in Python 3.10.
- def names a function. Return values default to None.
- Default arguments are evaluated once, at function definition.
- Use a parameter default of None when the default must be a fresh list or dict.

New enough to notice

- 3.14: t-strings (t'...') return a string.templatelib.Template, not a str.
- 3.14: annotations are stored and evaluated lazily (PEP 649 and PEP 749).
- 3.14: except A, B: is allowed when you do not use as.
- 3.15.0 was released on 9 October 2026. Highlights include lazy import, frozendict, sentinel, and unpacking in comprehensions. This lab's browser runtime is CPython 3.14.2, so those 3.15-only forms are shown for reading and are not sent to Run.